

University of Sargodha

BS 6th Term Examination 2024

Subject: Computer Science

Paper: Compiler Construction (CSCC-306)

Time Allowed: 02:30 Hours

Maximum Marks: 60

Note: Objective part is compulsory. Attempt any three questions from subjective part.

Objective Part (Compulsory)

- Q.1. Write short answers of the following in 2-3 lines of each on your answer sheet. (2*12)**
- What is the benefit of syntax tree in analysis synthesis model of compilation?
 - Mention some important properties of the intermediate code.
 - Define semantic analysis.
 - Differentiate between top down parsing and predictive parsing.
 - Differentiate between NFA and DFA with the help of one example.
 - What is the difference between compiler and interpreter?
 - Differentiate between lexeme and pattern with the help of example.
 - What is dead code elimination?
 - Define left recursion.
 - Differentiate between lexical analysis and parsing.
 - Define transition diagram.
 - Define syntax directed translation.

Subjective Part (3*12)

- Q.2. Explain top down parsing with the help of one example.**
- Q.3. Write the grammar that recognize the following source code.**
- ```
Void Main () {
 Int a, c;
 a = b
 c = d + e;
}
```
- Q.4. Write the three address code of the following program.**
- ```
Switch (a)  
{  
    case 1: x = y + z;  
    case 2: u = v + w;  
    default: p = q + r;  
}
```
- Q.5. Write a short note on Compiler Construction Tools.**
- Q.6. Optimize the following code.**
- ```
x = 1
y = a * b + 3
z = a * b + x + z + 2
```

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